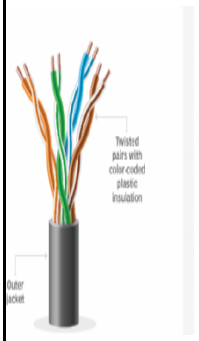
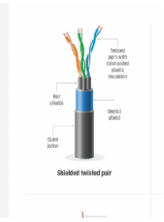
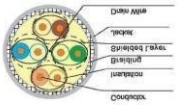
Ex.No.2	**Study of Network cables.**
Date:	1. Understand different types of Network cables. **2.** Make a cross-wired cable and straight through cable using clamping/crimping tool.

a. <u>Understand different types of network cable.</u>

Different type of cables used in networking are:

1. Unshielded Twisted Pair (UTP) Cable
2. Shielded Twisted Pair (STP) Cable
3. Coaxial Cable
4. Fibre Optic Cable

Cable type	**Category**	**Maximum Data Transmission**	**Advantages/ Disadvantages**	**Application/Use**	**Image**
UTP	Category 3	10 bps	**<u>Advantages</u>** • Cheaper in cost • Easy to install as they have a smaller overall diameter. **<u>Disadvantages</u>** • More prone to (EMI) Electromagnetic interference and noise	10Base-T Ethernet Fast Ethernet, Gigabit Ethernet Fast Ethernet, Gigabit Ethernet	
	Category 5	Up to 100 Mbps			
	Category 5e	1Gbps			

STP	Category6,6a	10Gbps	**Advantages** • Shielded. • Faster than UTP. • Less susceptible to noise and interference **Disadvantages** • Expensive • Greater installation effort	Gigabit Ethernet, 10G Ethernet (55m) Widely used in data centres	
SSTP	Category 7	10Gbps		Gigabit Ethernet, 10G Ethernet (100m)	 

Coaxial cable	RG-6 RG-59 RG-11	10-100Mbps	• High bandwidth • Immune to interference • Low loss bandwidth • Versatile • **Disadvantages** • Limited distance • Cost • Size is bulky	Speed of signal is 500m Television network High speed internet connections	

fibre optics cable	Single mode Multi mode	100Gbps	**Advantages** • High speed • High bandwidth • High security • Long distance **Disadvantages** • Expensive • Requires skilled installers	• Maximum distance of fibre optics cable is around 100meters	

b. <u>Make Your Own Ethernet Cross-Over Cable/ Straight cable</u>

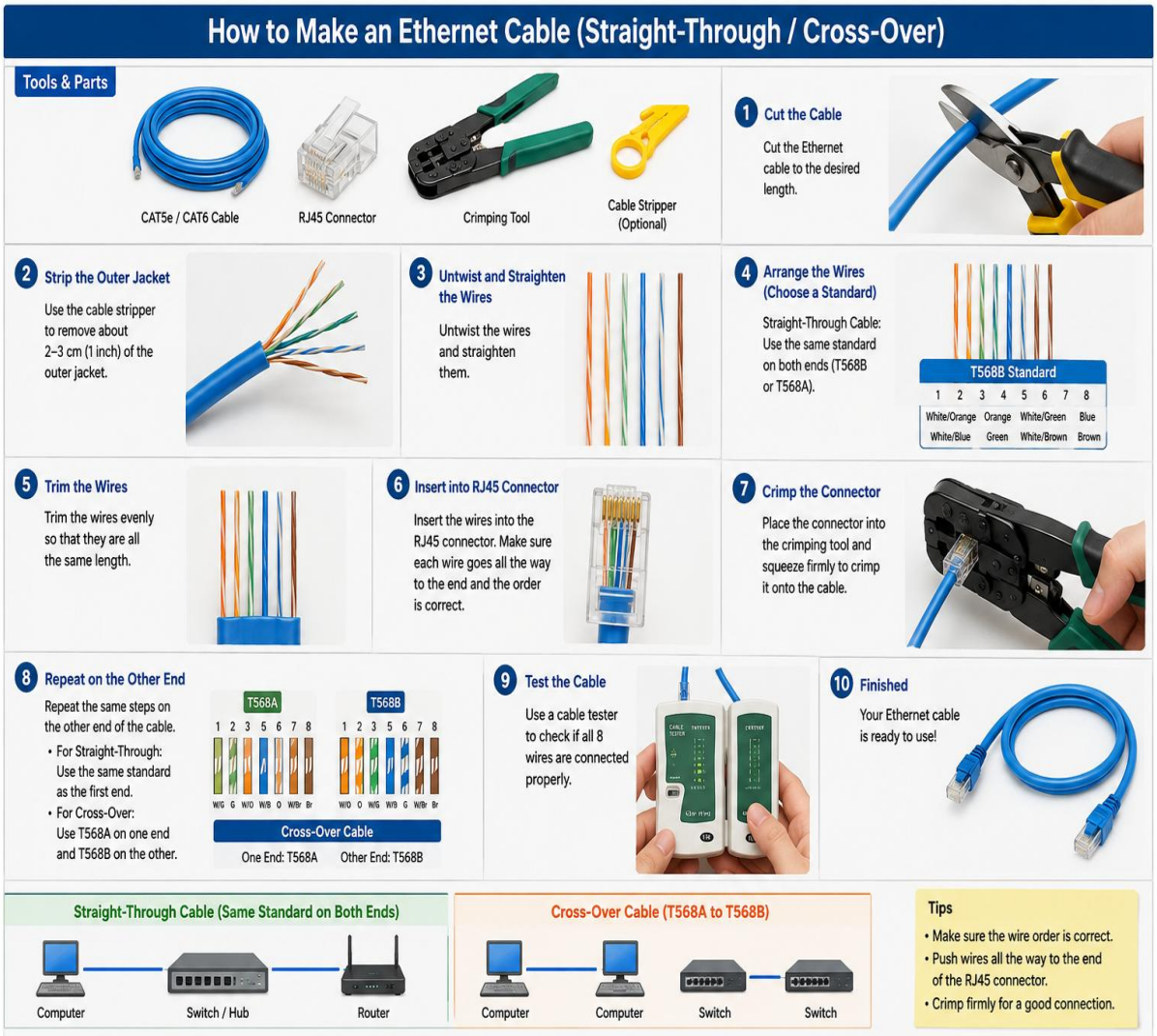
Tools and parts needed:

- Ethernet cabling. CAT5e is certified for gigabit support, but CAT5 cabling works as well, just over shorter distances.
- A crimping tool. This is an all-in-one networking tool shaped to push down the pins in the plug and strip and cut the shielding off the cables.
- Two RJ45 plugs.
- Optional two plug shields.

- **Ethernet Cable (CAT5e/CAT5)**
  - CAT5e is recommended because it supports Gigabit Ethernet (1000 Mbps).
  - CAT5 also works but supports lower speeds and shorter distances.
- **Crimping Tool**
  - A networking tool used to:
    - Strip the outer insulation of the cable.
    - Arrange and trim the wires.
    - Crimp (press) the RJ45 connector onto the cable.
- **RJ45 Connectors (2)**
  - Plastic connectors fitted at both ends of the Ethernet cable.

- Used to connect the cable to computers, switches, routers, etc.

- **Plug Shields (Optional)**
  - Protective covers placed over the RJ45 connectors.
  - Prevent damage to the locking clip and reduce strain on the cable.



RESULT:

The different types of Network cables were studied, and straight through and cross-over Ethernet cables were successfully made and tested using a crimping tool.